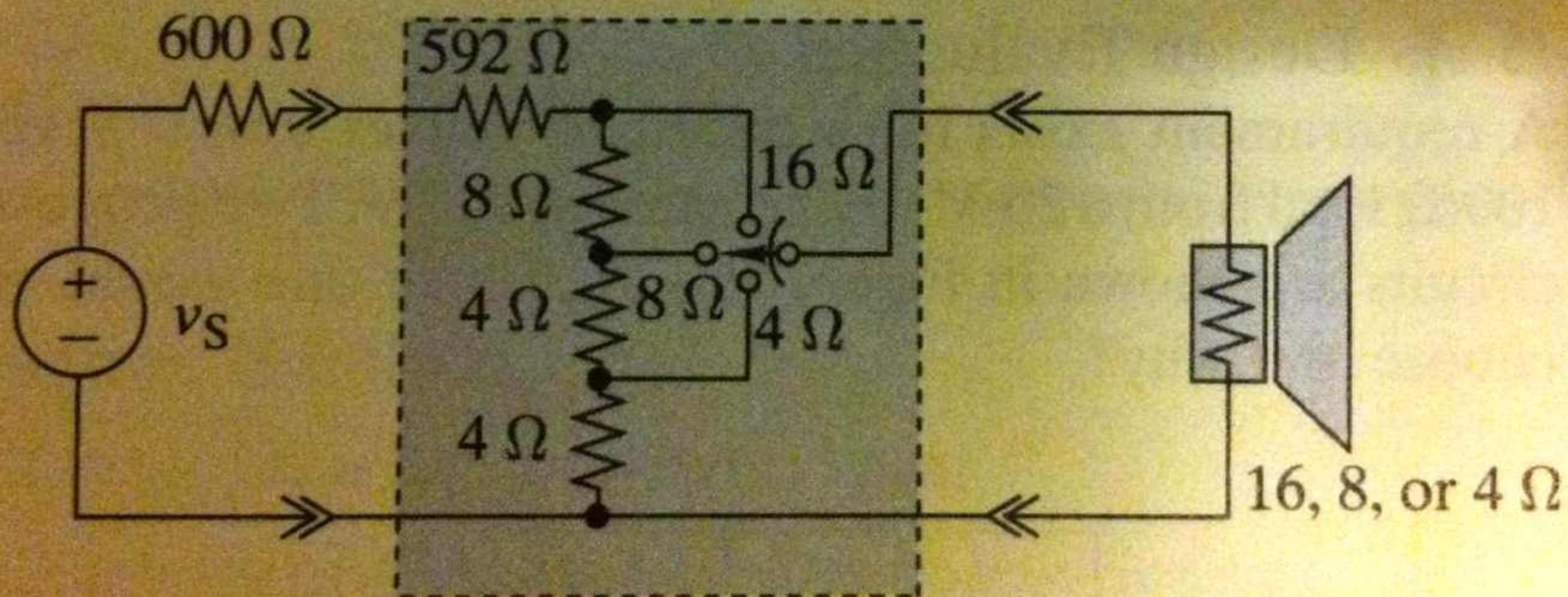


INTEGRATING PROBLEMS

3-71 **E** Audio Speaker Resistance-Matching Network

A company is producing an interface network that they claim will result in R_{IN} of $600\ \Omega \pm 2\%$ and R_{OUT} of 16, 8, or $4\ \Omega \pm 2\%$ —depending on whether the connected speakers are 16, 8, or $4\ \Omega$ —selectable via a built-in switch. Prove or disprove their claim.



Audio matching network

FIGURE P3-71

3-3 (a) Formulate node-voltage equations for the circuit in Figure P3-3.

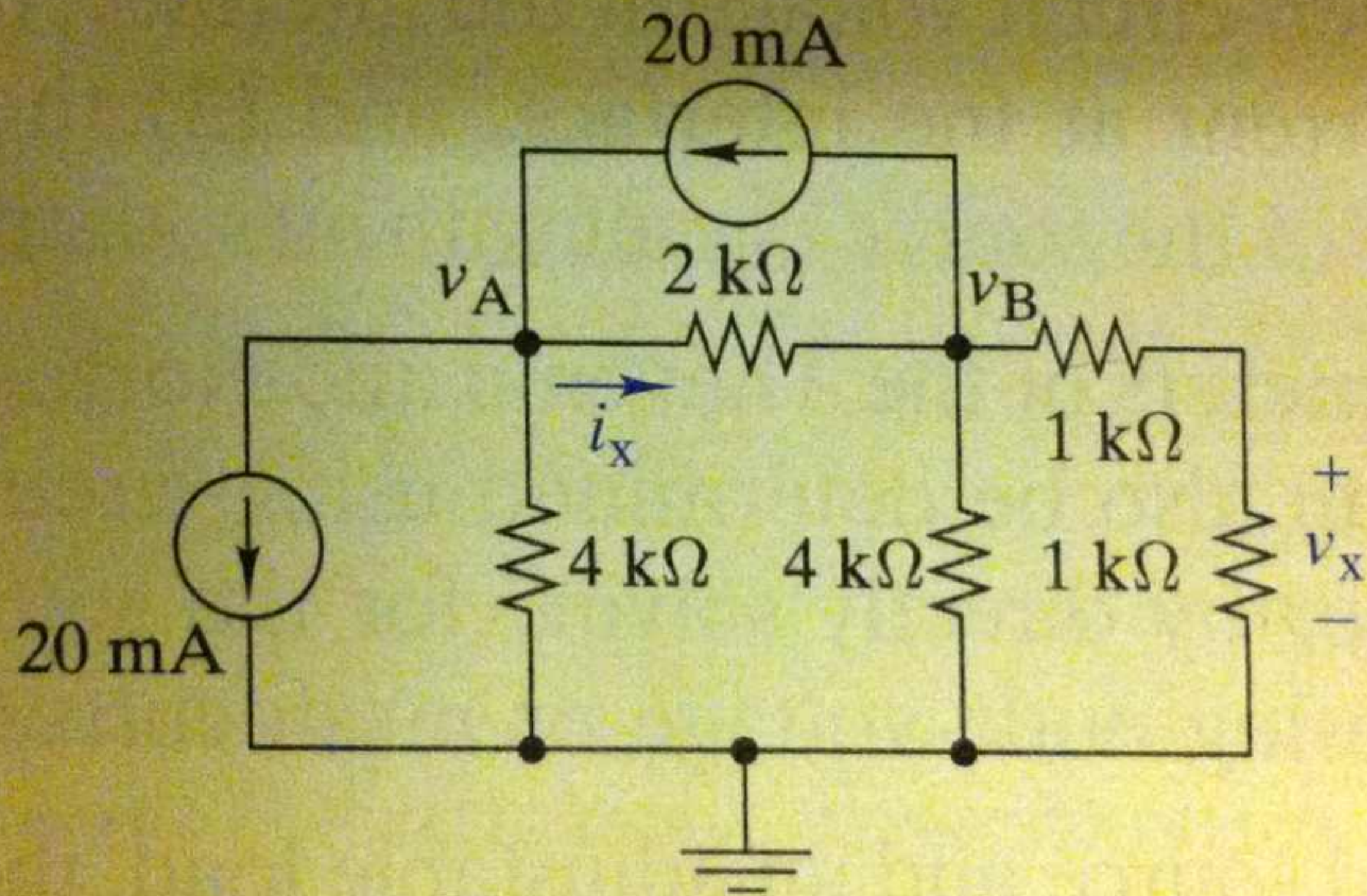


FIGURE P3-3

(b) Solve these equations for v_A and v_B .

(c) Use these results to find v_x and i_x .

- 3-5 (a) Choose a ground wisely and formulate node-voltage equations for the circuit in Figure P3-5.
- (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 1\text{ k}\Omega$, $v_S = 20\text{ V}$, and $i_S = 1\text{ mA}$.

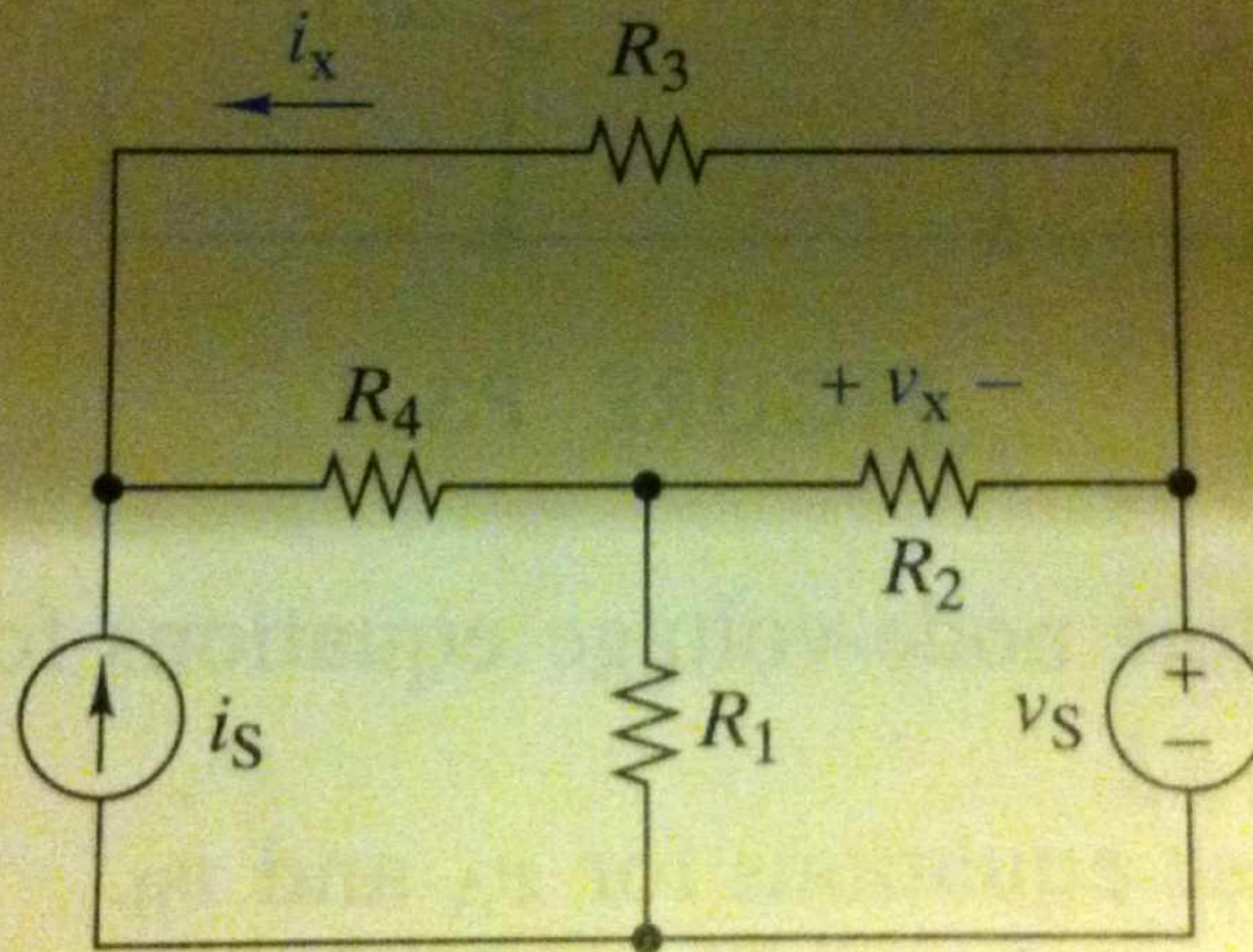


FIGURE P3-5

3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.

(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = 10\text{ k}\Omega$ and $v_S = 24\text{ V}$.

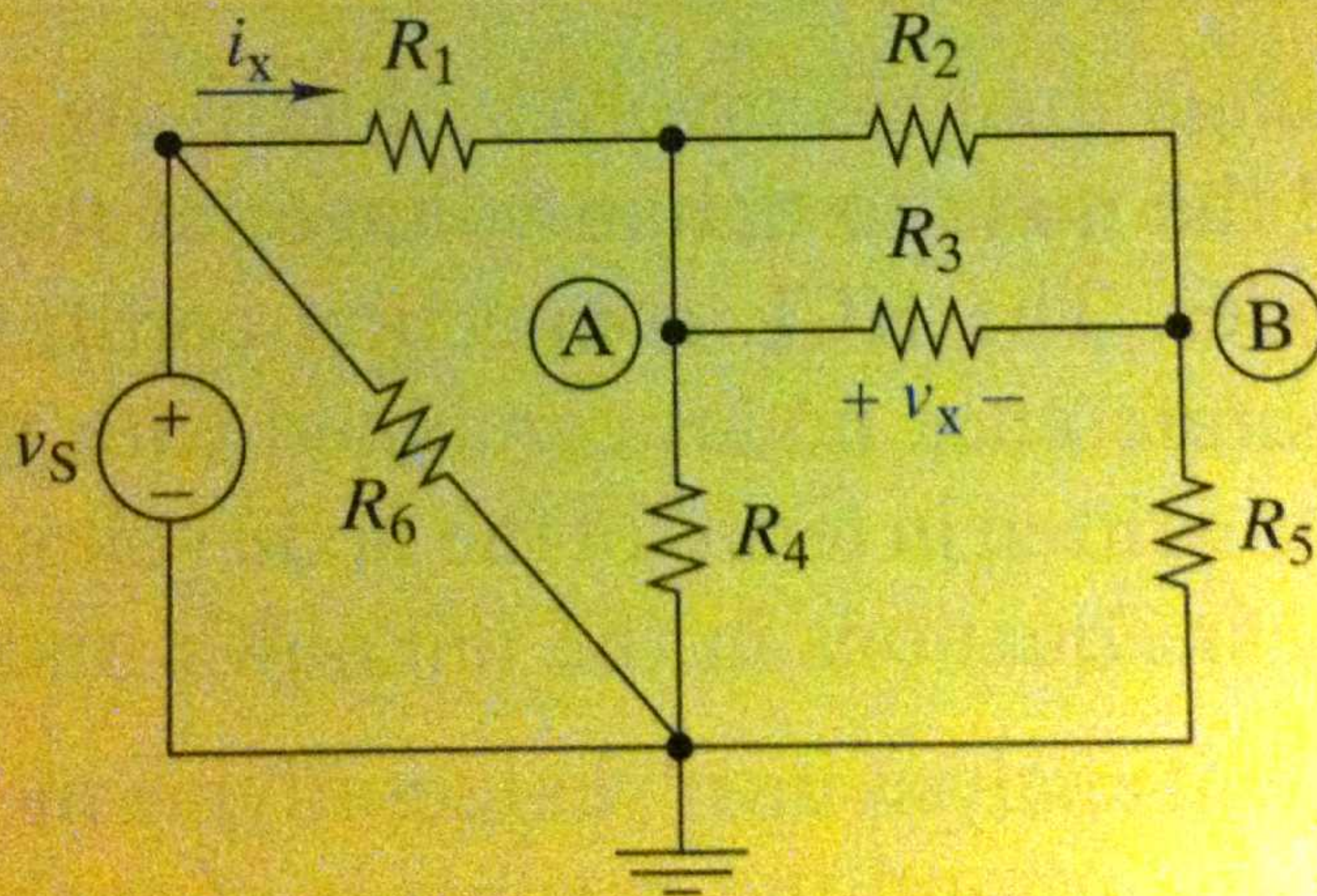


FIGURE P3-6

3-7 (a) Formulate node-voltage equations for the circuit in Figure P3-7.

(b) Use MATLAB to find symbolic expressions for the node voltages in terms of the parameters in the circuit.

(c) Find numeric values for v_A , v_B , and v_C when $R_1 = 1.5 \text{ k}\Omega$, $R_2 = 2.2 \text{ k}\Omega$, $R_3 = 3.3 \text{ k}\Omega$, $R_4 = 4.7 \text{ k}\Omega$, and $i_{S1} = i_{S2} = 2 \text{ mA}$.

(d) Use OrCAD to verify that your solution to part (c) is correct.

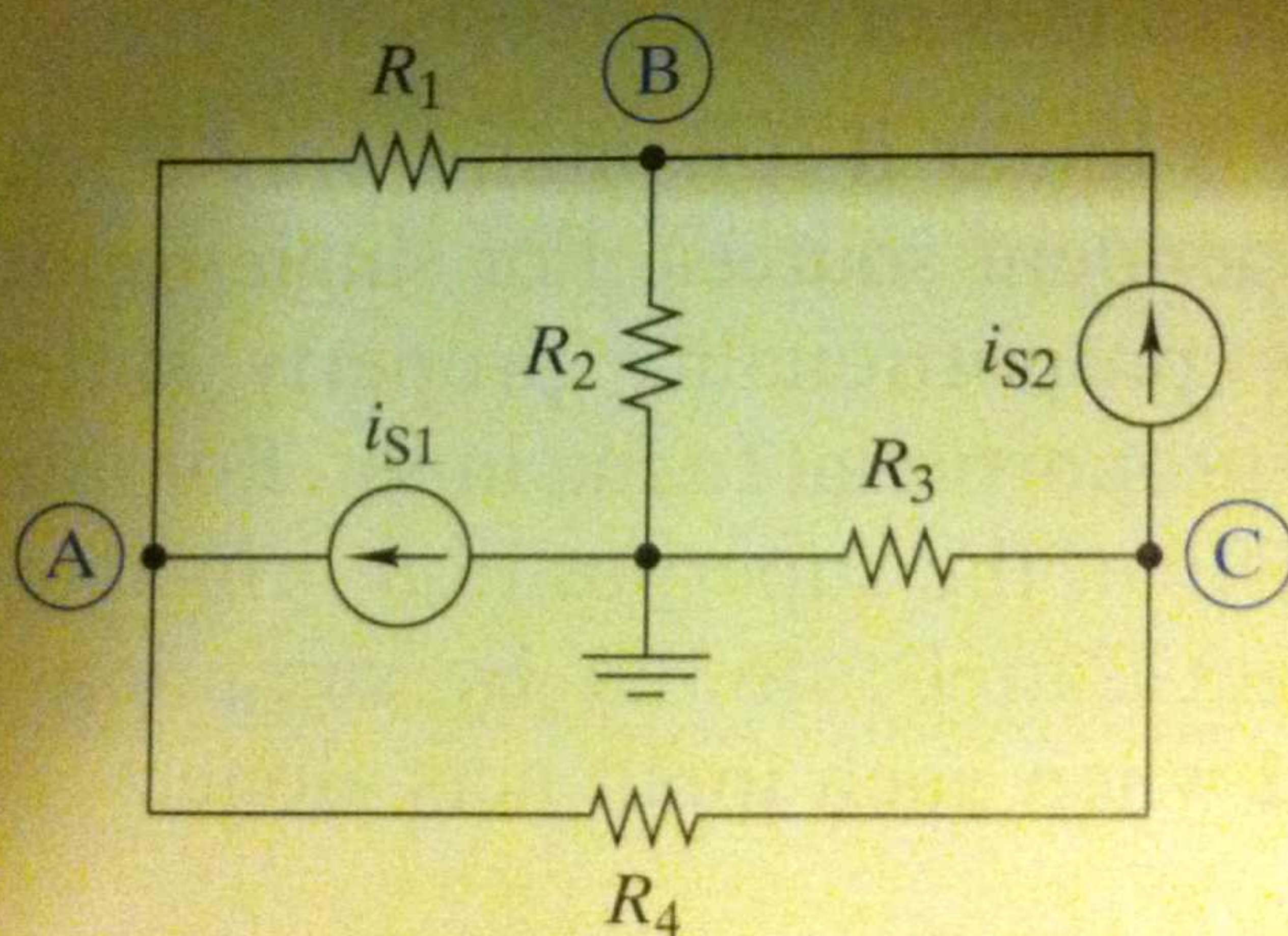


FIGURE P3-7

3-12(a) Formulate mesh-current equations for the circuit in Figure P3-12.

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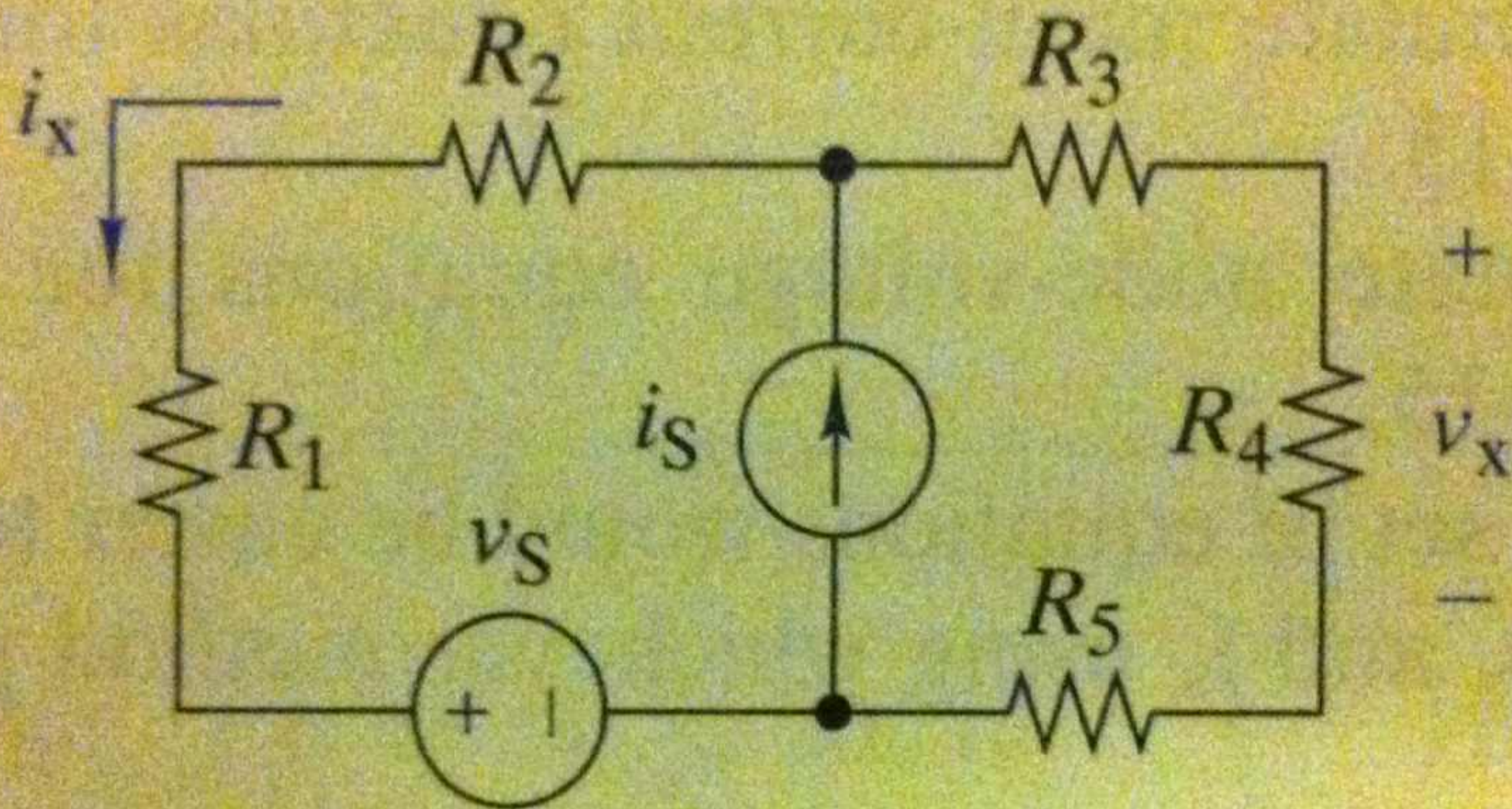


FIGURE P3-12

- (b)** Solve for v_x and i_x when $R_1 = 200\ \Omega$, $R_2 = 300\ \Omega$, $R_3 = 50\ \Omega$, $R_4 = 250\ \Omega$, $R_5 = 200\ \Omega$, $i_S = 100\ \text{mA}$, and $v_S = 25\ \text{V}$.
- (c)** Find the total power dissipated in the circuit.

- 3-20(a) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-20.
- (b) Solve for the same currents using node-voltage analysis and compare the effort required with each technique.
- (c) Use OrCAD to verify your results in the previous two parts.

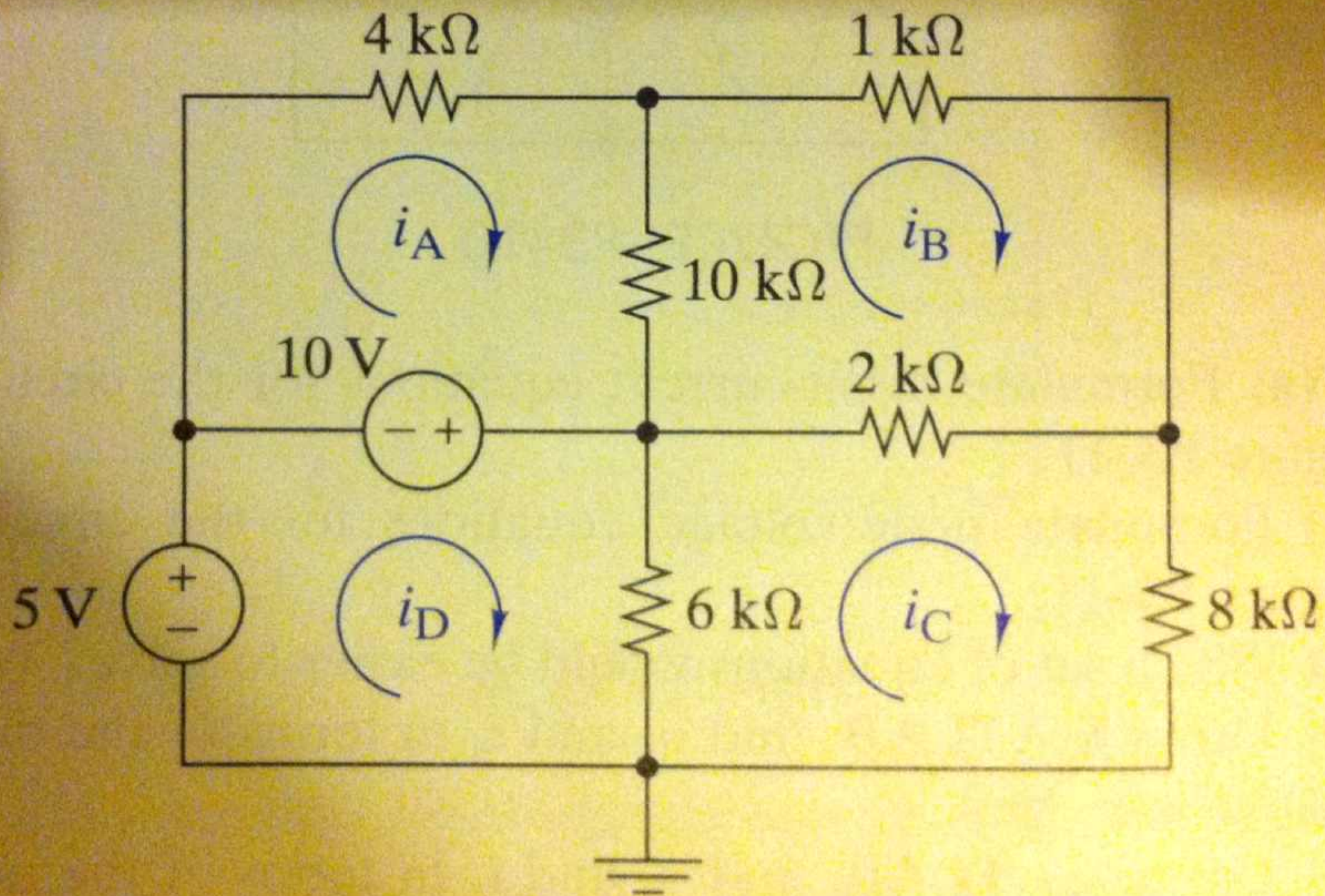


FIGURE P3-20